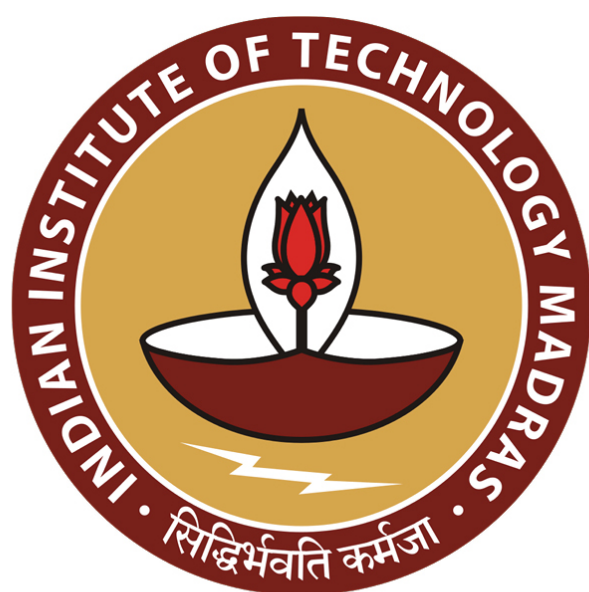


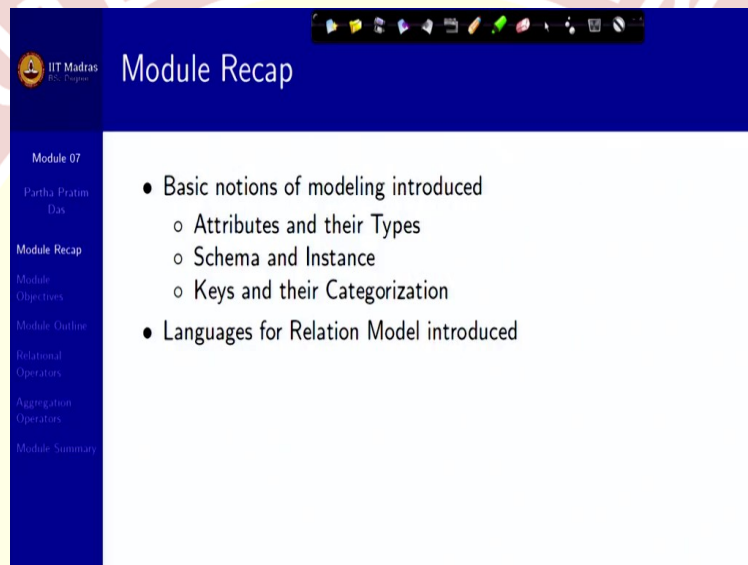


IIT Madras
ONLINE DEGREE

Database Management System
Professor. Partha Pratim Das
Department of Computer Science and Engineering
Indian Institute of Technology, Kharagpur
Introduction to Relational Model/2

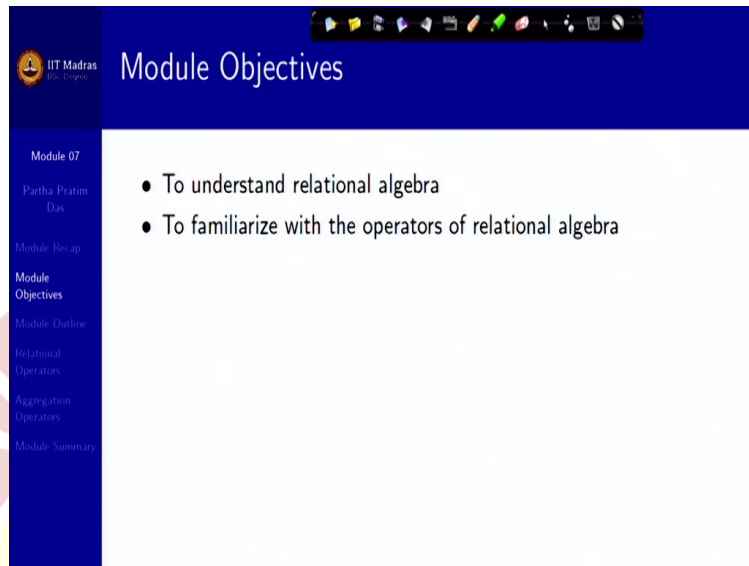
Welcome to module 7 of database management systems course in IIT Madras BSc online program.

(Refer Slide Time: 0:38)



In the last module we have got started to talk about specific details of relational model, in that we introduced the basic notions of attributes their types, relational schema and instance, and keys and their categorization, and we just get talked about a glimpse of the languages that relational algebra will relational system will have.

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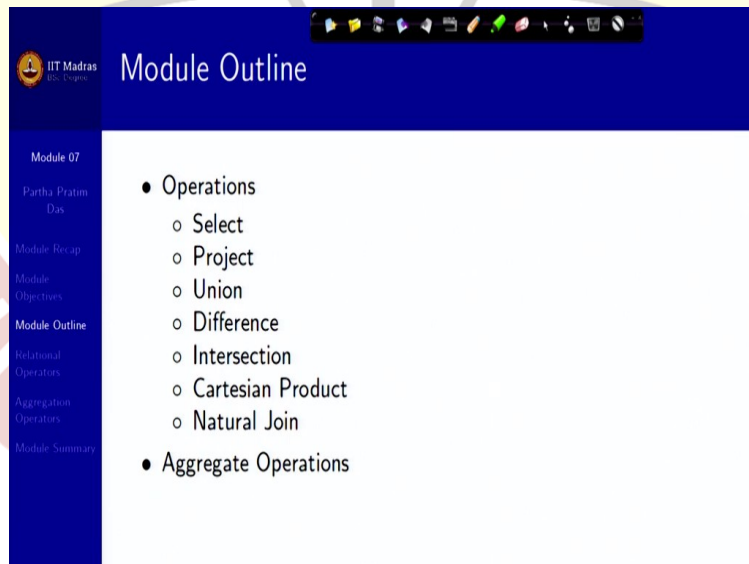


Module Objectives

- To understand relational algebra
- To familiarize with the operators of relational algebra

We will continue on that in this module and we will try to understand the relational algebra, the first language for handling query in greater depth particularly we talk about its operators in detail.

(Refer Slide Time: 1:18)



Module Outline

- Operations
 - Select
 - Project
 - Union
 - Difference
 - Intersection
 - Cartesian Product
 - Natural Join
- Aggregate Operations

So, these are the list of operators that we will specifically discuss one by one. So, let us get started.

(Refer Slide Time: 1:26)

The image shows two slides from a presentation. The top slide is titled "Relational Operators" and the bottom slide is titled "Basic Properties of Relations". Both slides have a blue header with the IIT Madras logo and a sidebar on the left with a table of contents. The bottom slide contains two bullet points and two tables illustrating the properties of relations.

Slide 1: Relational Operators

Module 07
Partha Pratim Das
Module Recap
Module Objectives
Module Outline
Relational Operators
Aggregation Operators
Module Summary

Relational Operators

Slide 2: Basic Properties of Relations

Module 07
Partha Pratim Das
Module Recap
Module Objectives
Module Outline
Relational Operators
Aggregation Operators
Module Summary

- A relation is set. Hence,
- Ordering of rows / tuples is inconsequential

A	B
a1	b1
a1	b2
a2	b1
a2	b2

is same as:

A	B
a1	b1
a2	b1
a2	b2
a1	b2

- All rows / tuples must be distinct

A	B
a1	b1

A	B
---	---

So, we talk about relational operators. Now, before I start talking of the operators I would like to remind you of something that I have already mentioned earlier is that, remember always that a relation is a set, so the ordering of elements or the ordering of rows and tuples in a relation is inconsequential, I can order them in whatever way I want.

So, if you look at the relation on left having two attributes A, B and 4 different tuples a1 b1, a1 b2, a2 b1, a2 b2, on the right you have them ordered in a different way, but it is the same set, so they are same, these relations are same, they are these instances are equivalent these instances are identical.

Second, a set must always have its elements as unique, so all rows and tuples must be distinct. So, on left I have a relational instance where a1 b1 is duplicated, so is a1 b2, these are not allowed. So, only one of those will have to be there, so what we show on right a1 b1 and a1 b2 is the correct relation the left one is wrong relation it will never exist. So, keep these two in mind as we discussed operators these principles will regularly periodically come into effect.

(Refer Slide Time: 3:11)

Select Operation – selection of rows (tuples)

Module 07
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Module Recap
Module Objectives
Module Outline
Relational Operators
Aggregation Operators
Module Summary

- Relation r

A	B	C	D
α	α	1	7
α	β	5	7
β	β	12	3
β	β	23	10

- $\sigma_{A=B \wedge D > 5}(r)$

A	B	C	D
α	α	1	7

Select Operation – selection of rows (tuples)

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- Relation r

A	B	C	D
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β	β	12	3
β	β	23	10

- $\sigma_{A=B \wedge D > 5}(r)$

A	B	C	D
α	α	1	7

$A = B$
 $D > 5$

Select Operation – selection of rows (tuples)

Module 07
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- Relation r

A	B	C	D
α	α	1	7
α	β	5	7
β	β	12	3
β	β	23	10

- $\sigma_{A=B \wedge D > 5}(r)$

A	B	C	D
α	α	1	7

Handwritten notes: A = B, A = B ✓, A = B ✓

So, we talk on different operators, first is called the select, and as the name suggest, select operation is to select a subset of rows from a relational instance which satisfy a particular condition. So, if you look at the select is written in terms of this notation, you know Greek symbol, and then it has as a subscript a Boolean condition, which is a condition that every tuple being selected from the original relational instance must satisfy and this is the name of the relational instance. So, this is your select is.

So, what does it say? It says $A = B$, this is the original relation r , that is I will take only those tuples where A is equal to B , that the first part of the condition, and then say and $D > 5$, so even within that I will take only those where $D > 5$. So, let us see let us try A equal to B , so these two will have to be identical, so this who is a candidate, this is a candidate, this is a candidate, this is crossed out because $\alpha \neq \beta$ A is not equal to B .

So, out of these three that we have got from A equal to B , since I have a conjunction, since I have and I have to check if $D > 5$, D is greater than 5, D is not greater than 5, cancel out, $D > 5$. So, finally I am left with two tuples, this and this, which is my selection result. What would have happened if instead of this if I would have or, instead of this if I would have or, then if I have $A = B$ or $D > 5$, A equal to B will take this, take this, take this, D greater than 5 will take this, take this and take this, and it is or, any one of them condition to satisfy.

So, if I do a select of A equal to B or D greater than 5 of r , then I will get the whole r itself, that this possible, but here it is and, so I got a smaller relation, so that is a selection is a very very

important, you can select one specific tuple out or you can take a set of tuples out. Let us, go to the next operation.

(Refer Slide Time: 6:21)

Project Operation – selection of columns (Attributes)

- Relation r

A	B	C
α	10	1
α	20	1
β	30	1
β	40	2

$\pi_{A,C}(r)$

A	C
α	1

A	C
α	1

Project Operation – selection of columns (Attributes)

- Relation r

A	B	C
α	10	1
α	20	1
β	30	1
β	40	2

$\pi_{A,C}(r)$

A	C
α	1

A	C
α	1

$\pi_{A,B}(r)$

Project Operation – selection of columns (Attributes)

Module 07
Partha Pratim Das
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Module Summary

- Relation r

A	B	C
α	10	1
α	20	1
β	30	1
β	40	2

$\pi_{A,C}(r)$

A	C
α	1

A	C
α	1

Handwritten notes: $\pi_{A,C}$, $= ?$

Project Operation – selection of columns (Attributes)

Module 07
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Module Summary

- Relation r

A	B	C
α	10	1
α	20	1
β	30	1
β	40	2

$\pi_{A,C}(r)$

A	C
α	1

A	C
α	1

Handwritten notes: $\pi_{A,C}$, $= \pi_{A,C}$

The next operation is called projection, that is, it selection was taking out few rows horizontally, projection will take out some columns and if you take out a column you take out the column as a whole. So, projection is written in terms of by pi, pi here and you would list the columns that you want to project out, that you want to keep, anything that you do not right here is taken out and then the original relational schema, so you have A, B, C and you are doing projection on A, C, so this B will be simply erased out.

If you erased out B then this is what you get, alpha 1, alpha 1, beta 1, beta 2. Now, remember that principle that every row in a relation has to be unique. So, I cannot have entries like this, so one of them will get deleted and what I get is this relation. What will happen if I do $\pi_{A,B} r$, say

pi A, B of r, what do I get? This is gone, now I will get this and you can see that all of them incidentally are unique, so there will be no removal of row that we will happen, what will be pi A of r? Only alpha and beta, so it is A, so that is the basic principle of projection. So, we know horizontal cuts, we know vertical cuts. Let us, go to the next one.

(Refer Slide Time: 8:39)

The slide is titled "Union of two relations" and is from IIT Madras. It shows two relations, r and s , and their union $r \cup s$.

Relation r :

A	B
α	1
α	2
β	1

Relation s :

A	B
α	2
β	3

The union $r \cup s$ is shown below:

A	B
α	1
α	2
β	1
β	3

This is more classical set theory operation that is union. So, if two relations have identical set of attributes, then you can take their relation, if they take their union, I mean obviously if this is AB attribute AB and that is attribute CD you cannot do any union, AB and AB. If you are a union what will happen?

Then all tuples, it is just that after this table you put the rest, and then unify unification is required, because here if you do union, you will find that this and this will come twice because we are putting them together. So, you will have to unify and have it only once, this is your union. Very simple. Next is difference.

(Refer Slide Time: 9:28)

Set difference of two relations

- Relation r, s

A	B
α	1
α	2
β	1

r

A	B
α	2
β	3

s

- $r - s$

A	B
α	1

There is a difference set, I have a set r , I have a set s , so this is my, what is difference? Differences what are there, what is there in r but not in s . So, you take out those things in s which are already in r . So, this is their, this is their, so you have to take out this element, this row and you are left with, so it is basically you can say that if you have, if you take the intersection of r and s and remove that from r then you have $r - s$, simple. But this again is set theoretic, other than select and project that we have seen so far the other two union and set difference a simple set operations.

(Refer Slide Time: 10:44)

Set intersection of two relations

- Relation r, s

A	B
α	1
α	2
β	1

r

A	B
α	2
β	3

s

- $r \cap s$

A	B
α	2

Now, you have intersection as well tuples which are common to both, now intersection and difference are not independent, because we know that one can be written in terms of the other. So, if we have difference, even without the intersection operator you have the intersection because it is this, very easy to see, so it is $r - s$ is this, this is r , this is s , may if you do $r - (r - s)$ that is from r if you take this out, then what will be left is this part which is intersection, so that is simple set algebra.

(Refer Slide Time: 11:28)

Joining two relations – Cartesian-product

- Relation r, s

A	B
α	1
β	2

r

C	D	E
α	10	a
β	10	a
β	20	b
γ	10	b

s

- $r \times s$

A	B	C	D	E
α	1	α	10	a
α	1	β	10	a
α	1	β	20	b
α	1	γ	10	b
β	2	α	10	a
β	2	β	10	a
β	2	β	20	b
β	2	γ	10	b

Handwritten notes: $r = \pi$, $s = \pi$

When you have Cartesian product again a set operation that is you know take the columns of this take the columns of this make all possible combinations. So, you have 2 tuples here, you have 4 tuples here, so you have 2 cross 4 is 8 tuples you will get, 1, 2, 3, 4, 5, 6, 7, 8. So, if you really see that all possible pairing of the tuples in table one and table two are present in the result which is the Cartesian product of the two relations.

And it is easy to see that if you take a Cartesian product, then you can get back, how do you get back the relation original relation? So, I can say that r is equal to we take Cartesian product of r and s and or rather I will write it as capital R , to mean that is a relational schema of R . So, if I take the Cartesian product and again project it on AB , I will get back to relation R . Similarly, for s , s is my new I mean I am using capital here to mean that R basically is A, B , S basically is C, D, E those are the schema, because what we have in r and s lower case are basically the instances. So, this is the Cartesian product which was a set.

(Refer Slide Time: 13:16)

The slide is titled "Cartesian-product – naming issue". It shows two relations, r and s , and their Cartesian product $r \times s$.

Relation r has attributes A and B .

A	B
α	1
β	2

Relation s has attributes B , D , and E .

B	D	E
α	10	a
β	10	a
β	20	b
γ	10	b

The Cartesian product $r \times s$ is shown below, with the attribute B from relation r qualified as $r.B$ and the attribute B from relation s qualified as $s.B$.

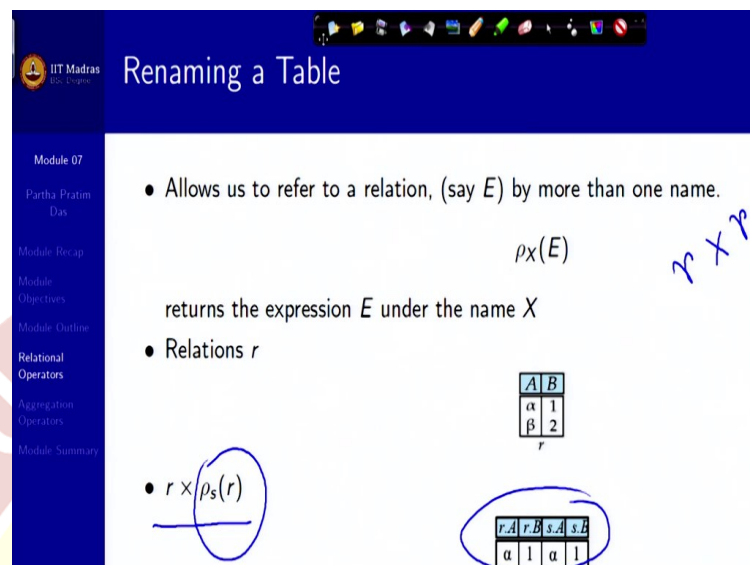
A	$r.B$	$s.B$	D	E
α	1	α	10	a
α	1	β	10	a
α	1	β	20	b
α	1	γ	10	b
β	2	α	10	a
β	2	β	10	a
β	2	β	20	b
β	2	γ	10	b

Now, you get into little nuances, because suppose between the two relations that you want to do Cartesian product with suppose there is a column one or more columns, which have the same name, now in the result relation you cannot have this because you cannot have two attributes by the same name, so what is done is very simple you just unify the name. How do you unify? You make this B as $r \cdot B$ and make this B as $s \cdot B$.

So, when you see that between the two relations there is one or more attribute which have which are identical, then you just qualify the name of that attribute with the name of the relation, this r and s are different, so their names are different. So, $r \cdot B$ and $s \cdot B$ are different. And then you do the normal Cartesian product.

So, only thing is you can you will have a different set of names, you will still have 5 attributes, but you will have different names for them. So, this is kind of a little bit additional trick that you need for the relational algebra, which you do not need in the set theory, because in set theory you do not talk about unification of attribute names.

(Refer Slide Time: 14:42)



Renaming a Table

- Allows us to refer to a relation, (say E) by more than one name.

$\rho_X(E)$

returns the expression E under the name X

- Relations r

A	B
α	1
β	2

r

$r \times \rho_s(r)$

$r.A$	$r.B$	$s.A$	$s.B$
α	1	α	1

Now, as you do that, two things one is how do you change this name? Is one question, other is maybe you have done this $r.B$ and $s.B$ and you are not happy you want to call $r.B$ as say p and $s.B$ as q , so how do you rename something? So, what you can say is you can and the renaming is also required because you may want to rename a relation because suppose you want to take you want to do a self-cross product r cross r .

Now, you see it bring a self-cross product you cannot use the earlier principle because A, B cross you have A, B , now by qualification $r.A$ and $r.A$, $r.B$ and $r.B$ they do not unify, because both relations are same. So, you can change the name of the relation, call it s , then the unification has happened, and in this way you can easily go around using any names.

(Refer Slide Time: 15:59)

Composition of Operations

- Can build expressions using multiple operations
- Example: $\sigma_{A=C}(r \times s)$
- $r \times s$

A	B	C	D	E
α	1	α	10	a
α	1	β	10	a
α	1	β	20	b
α	1	γ	10	b
β	2	α	10	a
β	2	β	10	a
β	2	β	20	b
β	2	γ	10	b

- $\sigma_{A=C}(r \times s)$

Now, naturally like the set theory, you can do you can compose any number of operations. For example, here what we are doing is we have relations r and s , we have done a this is a Cartesian product and then we have done a projection, I am sorry then we have done a selection A equal to s , so this is the result of the Cartesian product and then you say this, this, this, will survive because in those A equal to s , everything else is removed. So, you can do any number of composition of operators. So, it is like just building up the algebra.

(Refer Slide Time: 16:49)

Joining two relations - Natural Join

- Let r and s be relations on schemas R and S respectively. Then, the relations R and S is a relation on schema $R \cup S$ obtained as follows:
 - Consider each pair of tuples t_r from r and t_s from s .
 - If t_r and t_s have the same value on each of the attributes in $R \cap S$, then the result, where
 - ▷ t has the same value as t_r on r
 - ▷ t has the same value as t_s on s

Now, we get a new operation which is more specific for relational algebra, let us say R and S are relations on schemas capital R and capital S. The natural join, mind this word, the natural join, Cartesian product in general will also called join, natural join of R and S is a relation on the schema R union S, that is you have the attributes of R and you have the attributes of S, so that you take every pair of tuple from instance R and every tuple for instance S and so that is a kind of a Cartesian product, so you are making combinations, but there is more to it.

If t_r and t_s have the same value on each attribute in $R \cap S$, that is if there is one attribute which is common between R and S, then tuples which have identical value on that attribute will only be taken in the natural join, that is not true for Cartesian product, in Cartesian product to take all possible values.

And therefore, we if the for attributes that are in $R \cup S$ we are using the qualified them $r.A$, $s.A$, $r.B$, $s.B$, here no, if two attributes have identical name between R and S then you only keep those tuples which have identical value on those identical attributes, that is what it is saying, example.

(Refer Slide Time: 18:54)

The slide is titled "Natural Join Example" and is from IIT Madras. It shows two relations, R and S, and their natural join. Relation R has attributes A, B, C, D and Relation S has attributes B, D, E. The natural join of R and S is shown as a table with attributes A, B, C, D, E. Handwritten annotations include "RUS", "RNS", and "25" in a circle, indicating the result of the join.

• Relations r, s :

A	B	C	D
α	1	α	α
β	2	γ	α
γ	4	β	b
α	1	γ	a
δ	2	β	b

• Natural Join

o $r \bowtie s$

A	B	C	D	E
α	1	α	α	α
α	1	α	α	γ
α	1	γ	a	α
α	1	γ	a	γ

So, look at these carefully, this is r, this is s, so you know this is capital R and this is capital S, so $R \cup S$ is A, B, C, D, E; $R \cap S$ is B, D. So, I have two identical column names, B and B, D and D, so when I do the join, I have 5 records here, I have 5 records here, I should get 25 in the

Cartesian product, but here know no, when I pair with this tuple any tuple here I have to make sure that the B and D values are identical.

So, I will be able to pair this with this and with this, but not with this, that is the first record or first row on the left on R can be paired with the first row on S, because B have identical values D have identical values, it can be paired with the third because they have identical values, it can be not be paired with anyone else. So, you have this resulting from that pairing, the next one resulting from that pairing.

If you follow this all through is what you get the natural join on this. So, what will happen in natural join? $r.B$ and $s.B$ necessarily similarly $r.B$ and $s.D$ these two will necessarily have identical values in the columns because that is how you are choosing, 1 1 you have chosen, similarly let us say if you take the next one, 2a there is not 2a, so this entire thing will not produce anything, take the next one, you have 4b, there is no b here, entire thing will not have anything, then you have this which has 1a, so again you will have two of this, then you have delta which is 2b, so 2b will only have this, this right now.

Now, necessarily on both these $r.B$ and $s.B$ you will have identical values and on $r.D$ and $s.D$ will also have identical values, this is by the way of making the choice, so it is redundant to keep both, so you remove 1 from the table and once you remove you do not need to call it by a qualified name, so that is exactly what how you get this table, so, in this table you do not get 2B's or 2D's that are identical.

So, if you look at what you have done, you have done a cross product, then on identical column names $r.B$ and $s.B$ you have checked if the values are equal on B equal on D inclusive, they will have to be equal on both, you have selected only those tuples, so then you have got 7 columns of that you have taken the distinct 3 and of the duplicate ones you have taken just 1, removing the other two, you have 5. We will see that this is a very, very powerful operation which we will use rampantly, rampantly I reiterate across different query processing. So, you will have to practice and learn this really well.

(Refer Slide Time: 23:38)

The top screenshot shows a presentation slide titled "Aggregation Operators" from IIT Madras. The sidebar menu includes: Module 07, Partha Pratim Das, Module Recap, Module Objectives, Module Outline, Relational Operators, Aggregation Operators, and Module Summary.

The bottom screenshot shows the same slide with the following content:

- Can we compute:
 - SUM
 - AVG
 - MAX
 - MIN

Handwritten example:

A	B
a	5
b	2
c	3

Below the table, there is a handwritten expression: $\sigma_{B \geq 2}(\dots)$ with an arrow pointing to the table.

So, that was about the relational operators, now other than that there are some aggregate operators. So, relational operators you will have to remember that the operators always take one or two relations, some operators are unary, like select, like project, so they take one relation and give you one relation, some are binary like union, intersection, difference, Cartesian product, natural join, they take two relations and give you one relation.

Remember that the input will always have to be relation the output will always be a relation. And it is possible that in some cases you might get an empty relation also, relation which does not have any tuple, so we will go back and say there is no answer. Now, there are these are the basic

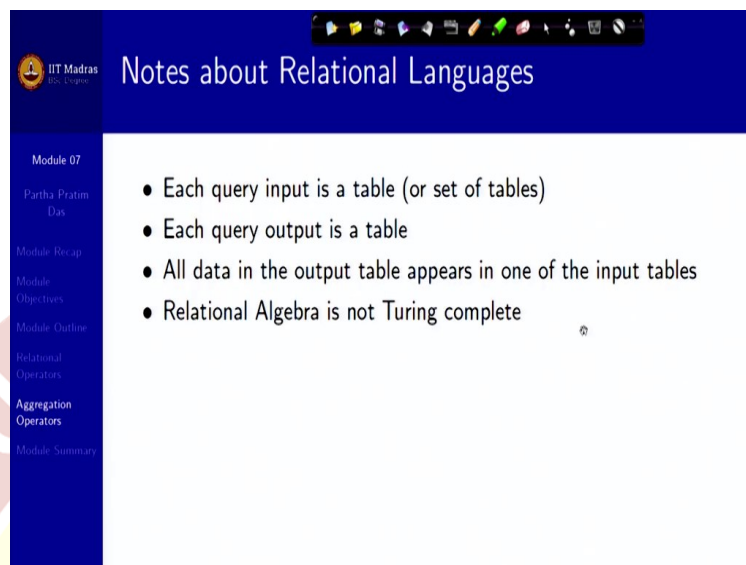
relational operators, then we have a set of what is called aggregate operators, the concept of the aggregate operators is that you take a relation again, then based on one or more columns or certain conditions, you do an aggregation, so kind of you have a table A, B, I say that I want to do an aggregation on column B, what is that aggregation? I want to make a sum.

Of course, the type of that column has to be such that the agree, that particular aggregate operation is possible. So, I can do a sum on this column to get 10, I can do a max on this column to get 5, like this, you can do a aggregation on a relationally operated file also, for example, I will say that well, I want to do is I want to aggregate values that are more than 2, so I will take this r and then do B greater than 2r, what does this give me? This gives me a new relation, relation which does not have this, and on this I do a sum on B to get 8.

So, two main differences between the relational operators and aggregation operators, relational operators are always dealing with certain structure tuples and so on and always will produce a relation, whereas aggregate operations will do an a typical aggregate operations a numerical as you can see all of these are numerical operators, they will necessarily take one or more columns, the tuples and do some aggregation arithmetic on those and they will produce a single value, they do not produce a table, settle, because the sum of your budget or your maximum salary of an employee or the average execution time of a query and so on these are the different aggregations that can happen, does not will not produce a multiple value.

So, in aggregation you input is a table and the output is a single value, so that is how these are distinguish, aggregation operators are not fundamental to pure relational algebra, because relational algebra only talks about relation, but these are as we will see, will come as additional operators as we write relational algebra in the form of SQL kind of language because they become very handy and we will show you how to do those in time.

(Refer Slide Time: 28:40)

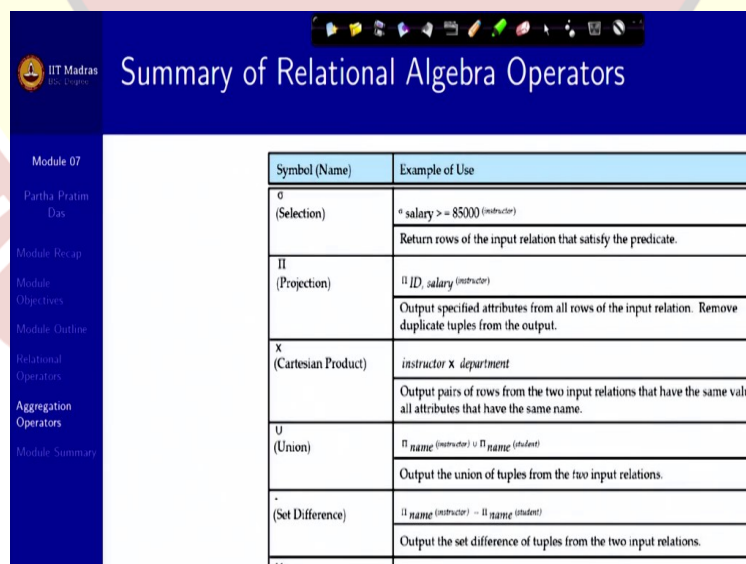


Notes about Relational Languages

- Each query input is a table (or set of tables)
- Each query output is a table
- All data in the output table appears in one of the input tables
- Relational Algebra is not Turing complete

So, that was, so each query input is a table and a set of tables and each query output is a table. Each data in the output table appears in one of the input tables. Relational algebra is not Turing complete, which means there are problems there are programs for which no relational algebraic solution can be read. And that is one of the reasons that you will necessarily have to go with host language.

(Refer Slide Time: 29:14)

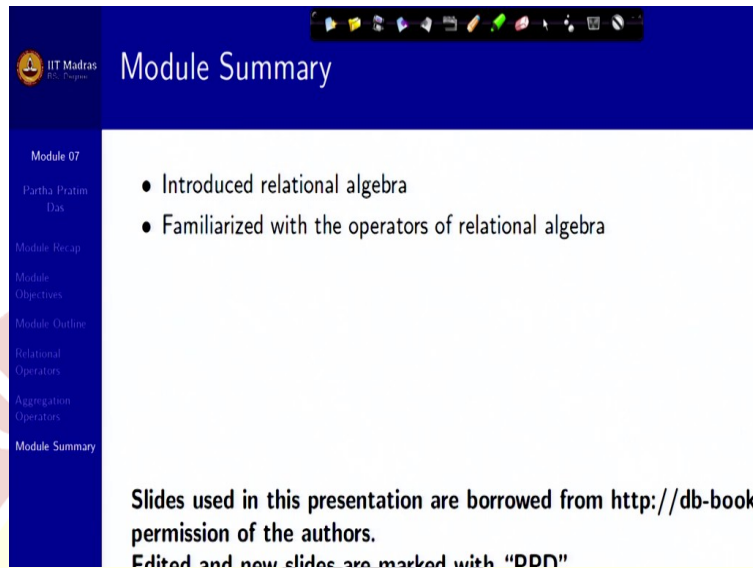


Summary of Relational Algebra Operators

Symbol (Name)	Example of Use
σ (Selection)	$\sigma^s \text{ salary} >= 85000 (\text{instructor})$ Return rows of the input relation that satisfy the predicate.
π (Projection)	$\pi ID, salary (\text{instructor})$ Output specified attributes from all rows of the input relation. Remove duplicate tuples from the output.
$\times$ (Cartesian Product)	$\text{instructor} \times \text{department}$ Output pairs of rows from the two input relations that have the same value all attributes that have the same name.
$\cup$ (Union)	$\pi \text{ name} (\text{instructor}) \cup \pi \text{ name} (\text{student})$ Output the union of tuples from the two input relations.
$-$ (Set Difference)	$\pi \text{ name} (\text{instructor}) - \pi \text{ name} (\text{student})$ Output the set difference of tuples from the two input relations.

So, these are summary, this is just for your quick reference till you get totally comfortable with these operators which we will have to soon get comfortable with.

(Refer Slide Time: 29:25)



Module Summary

- Introduced relational algebra
- Familiarized with the operators of relational algebra

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So, in the module today we have introduced relational algebra with its basic operators so on, thank you very much for your attention and we will continue on this in the next module. Have a nice day.

